

Generative AI Capstone Project Completion Guide

This guide provides a structured approach for planning, building, documenting, and presenting a Generative AI capstone project.

1. Project Selection

- Identify a real business or user problem.
- Choose a Generative AI use case such as chatbot, document summarization, content generation, knowledge assistant, code assistant, or recommendation system.
- Define project objectives, scope, and success metrics.

2. Requirement Analysis

- Define target users and stakeholders.
- Collect sample data and reference materials.
- Document functional and non-functional requirements.

3. Solution Design

- Create system architecture diagram.
- Select foundation model (GPT, Llama, Mistral, etc.).
- Decide whether RAG, fine-tuning, prompt engineering, or agents will be used.

4. Development

- Prepare development environment.
- Build data ingestion and preprocessing pipeline.
- Implement prompts and model interactions.
- Develop user interface or API.

5. Testing and Evaluation

- Test accuracy, relevance, latency, and reliability.
- Measure performance using defined metrics.
- Document limitations and risks.

6. Documentation

- Prepare project report including problem statement, methodology, architecture, implementation, results, and future enhancements.
- Include screenshots, diagrams, and evaluation results.

7. Final Presentation

- Create 10-15 slide presentation.
- Demonstrate live application or recorded demo.
- Explain business value, technical implementation, and lessons learned.

Recommended Deliverables

- Source Code Repository
- Project Report
- Architecture Diagram
- Demo Video
- Presentation Deck
- Test Results and Evaluation Summary